Computer Science – the Year 10 Record

OCR J277 • every lesson recorded last year, in the order it was taught

Computer Science 11 • papers summer 2027

This is the record of your Year 10, taken from the class notebook: 37 dated pages between August 2025 and May 2026. Not every lesson leaves a page behind, so this is not the whole story of your year — but it is an accurate account of what was written down, and it is where we start from. The right-hand column gives the **OCR specification reference**, so you can match any lesson to the syllabus itself. Page 2 turns the same information into a map of the whole course.

Date	Day	Lesson	OCR ref
Autumn 2025 — Data representation, then hardware			
21 Aug 2025	Thu	Denary (decimal) and binary	1.2.4
2 Sep 2025	Tue	Hexadecimal	1.2.4
5 Sep 2025	Fri	Adding binary and binary shifts	1.2.4
9 Sep 2025	Tue	Character sets	1.2.4
12 Sep 2025	Fri	File sizes	1.2.3
16 Sep 2025	Tue	Representing images	1.2.4
19 Sep 2025	Fri	More image representation	1.2.4
26 Sep 2025	Fri	Sound recording	1.2.4
30 Sep 2025	Tue	Data compression	1.2.5
3 Oct 2025	Fri	Questions	1.2
		— October break —	
28 Oct 2025	Tue	Introduction to the CPU	1.1.1
18 Nov 2025	Tue	Embedded systems	1.1.3
18 Nov 2025	Tue	Questions on CPU and embedded systems	1.1.1–3
21 Nov 2025	Fri	CPU performance, memory and RAM	1.1.2
21 Nov 2025	Fri	RAM and ROM — primary storage	1.2.1
25 Nov 2025	Tue	Secondary storage	1.2.2
		— Christmas —	
Spring 2026 — Boolean logic, then networks			
6 Jan 2026	Tue	Introduction to Boolean logic	2.4.1
13 Jan 2026	Tue	Boolean logic, part 2	2.4.1
16 Jan 2026	Fri	AND, OR and NOT	2.4.1
23 Jan 2026	Fri	Network questions	1.3.1
27 Jan 2026	Tue	Client–server and peer-to-peer	1.3.1
30 Jan 2026	Fri	Networks 3	1.3.1
3 Feb 2026	Tue	Networks 4 — performance	1.3.1
6 Feb 2026	Fri	Network issues	1.3.1
6 Feb 2026	Fri	Network topologies	1.3.1
16 Feb 2026	Mon	Network protocols	1.3.2
20 Feb 2026	Fri	TCP/IP	1.3.2
24 Feb 2026	Tue	The concept of layers	1.3.2
27 Feb 2026	Fri	The internet	1.3.1
Spring–summer 2026 — network security, with some programming			
6 Mar 2026	Fri	Cyber security and social engineering	1.4.1
		— spring break —	
24 Mar 2026	Tue	Shoulder surfing	1.4.1
24 Mar 2026	Tue	Malicious code	1.4.1
27 Mar 2026	Fri	Trace tables	2.1.2
31 Mar 2026	Tue	Viruses and malicious code	1.4.1
5 May 2026	Tue	Paper 2 questions	2.1–2.2
11 May 2026	Mon	Hacking	1.4.1
19 May 2026	Tue	Printout	2.2

The whole course, and where we are in it

Every topic on the OCR J277 specification is below, in specification order. ■ means it appears in last year's record. ⊡ means part of it does. ☐ means it is still ahead of us — that is simply the work of this year, and it is what Year 11 is for. Use this as a map: when you revise, revise by these numbers, because the exam is written to them.

Ref	Topic	Year 10
Paper 1 — Computer systems (50% of the qualification)		
1.1.1	Architecture of the CPU — fetch–execute, registers, von Neumann	■
1.1.2	CPU performance — clock speed, cores, cache	■
1.1.3	Embedded systems	■
1.2.1	Primary storage — RAM and ROM	■
1.2.2	Secondary storage — magnetic, optical, solid state	■
1.2.3	Units and file sizes	■
1.2.4	Data storage — binary, hexadecimal, characters, images, sound	■
1.2.5	Compression	■
1.3.1	Networks and topologies	■
1.3.2	Wired and wireless networks, protocols and layers	■
1.4.1	Threats to computer systems and networks	■
1.4.2	Identifying and preventing vulnerabilities	☐
1.5.1	Operating systems	☐
1.5.2	Utility software	☐
1.6.1	Ethical, legal, cultural and environmental impact	☐
Paper 2 — Computational thinking, algorithms and programming (50%)		
2.1.1	Computational thinking — abstraction, decomposition	☐
2.1.2	Designing, creating and refining algorithms — flowcharts, trace tables	⊡
2.1.3	Searching and sorting algorithms	☐
2.2.1	Programming fundamentals — variables, sequence, selection, iteration	☐
2.2.2	Data types	☐
2.2.3	Additional programming techniques — arrays, sub-programs, files, SQL	☐
2.3.1	Defensive design	☐
2.3.2	Testing	☐
2.4.1	Boolean logic — AND, OR, NOT, truth tables	■
2.5.1	Programming languages — high and low level, translators	☐
2.5.2	The Integrated Development Environment (IDE)	☐

What this means for our year — read this bit properly

- **Paper 1 is in good shape.** Eleven of its fifteen topics are already behind you. What is left there is four topics, and they are among the most straightforward on the course.
- **Paper 2 is where the year goes.** It is worth exactly as much as Paper 1, and most of it is still ahead. **Thirty of its eighty marks come from writing actual code** — so programming is not one topic among many, it is the largest single block of marks in the whole qualification.
- **So we start programming early, and we keep doing it all year.** Not a unit that finishes; a thing we do most weeks, alongside the Paper 1 topics that are left.
- **Nothing here is a verdict on you.** A course gets taught in the order someone chose, and this is simply where that order got to. You have a full year, and the map above is the plan.